

SOC-MartNet §4.1–4.8 Full Numerical Reproduction Report

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1. Notation and accuracy metrics

The notation below follows the conventions of paper §4. Networks: $u_\alpha, v_\theta, \rho_\eta$ denote the control / value / adversarial networks; W_h is the hidden width (default 6 hidden layers of $d + 10$ ReLU units; table shorthand $w_{256} = W_h = 256$, $w_{d+10} = W_h = d + 10$); r is the output dimension of ρ_η (default 600). Training: I is the number of iterations; N the number of time steps (default 100); T the terminal time (default 1); M the total number of sample paths (paper default 10^5 ; this reproduction uses 10^4 , with 6×10^3 for a few high-dimensional cells — see the parameter settings of each section); batch size $|M_i|$ defaults to 256 ($d \leq 1000$) / 128 ($d > 1000$); optimizer RMSProp; learning rates

$$\delta_1 = \delta_2 = \delta_0 \times 10^{-3} \times 0.01^{i/I}, \quad \delta_3 = 10^{-2} \times 0.01^{i/I}, \quad i = 0, 1, \dots, I-1,$$

where $\delta_0 = 3d^{-0.5}$ ($d \leq 1000$) and $3d^{-0.8}$ ($d > 1000$); λ is initialized at 10 with learning rate $\delta_4 = 10$ and upper bound $\bar{\lambda}$; $J = 2K = 2$. Throughout this report $\text{lr}_u / \text{lr}_\rho$ abbreviate the levels of $\delta_1 = \delta_2 / \delta_3$.

Error (paper definition):

$$\text{RE} := \left(\sum_{x \in D_0} |\hat{v}(0, x) - v(0, x)| \right) / \left(\sum_{x \in D_0} |v(0, x)| \right)$$

that is, the relative L^1 error (paper Table 1 uses the single-point convention $D_0 = \{x_0\}$; §4.3 of the paper denotes it RE_1); the relative L^∞ error is defined as

$$\max_{x \in D_0} |\hat{v}(0, x) - v(0, x)| / \max_{x \in D_0} |v(0, x)|.$$

This report uses final-iteration values; SD denotes standard deviation; in the history panels Mart.

Loss = $|G|^2$ and Hamilt. follows the paper definition. Reference solutions are the analytic or high-accuracy reference solutions of each problem (the top row of §4.2 uses a Cole–Hopf 10^6 -sample reference; §4.7 uses the $\varepsilon = 0$ reference solution).

2. §4.1 Linear parabolic equation (paper Fig. 1)

d	100	1000	2000
RE (5-seed)	$6.16\text{e-}3 \pm 1.24\text{e-}3$	$6.54\text{e-}3 \pm 0.68\text{e-}3$	$4.54\text{e-}3 \pm 0.57\text{e-}3$

Hardware: single RTX 4090 GPU.

This subsection of the paper reports illustrative results (no printed numeric anchor). All 12 panels (solution curves + Mart loss / RE histories) are rendered:

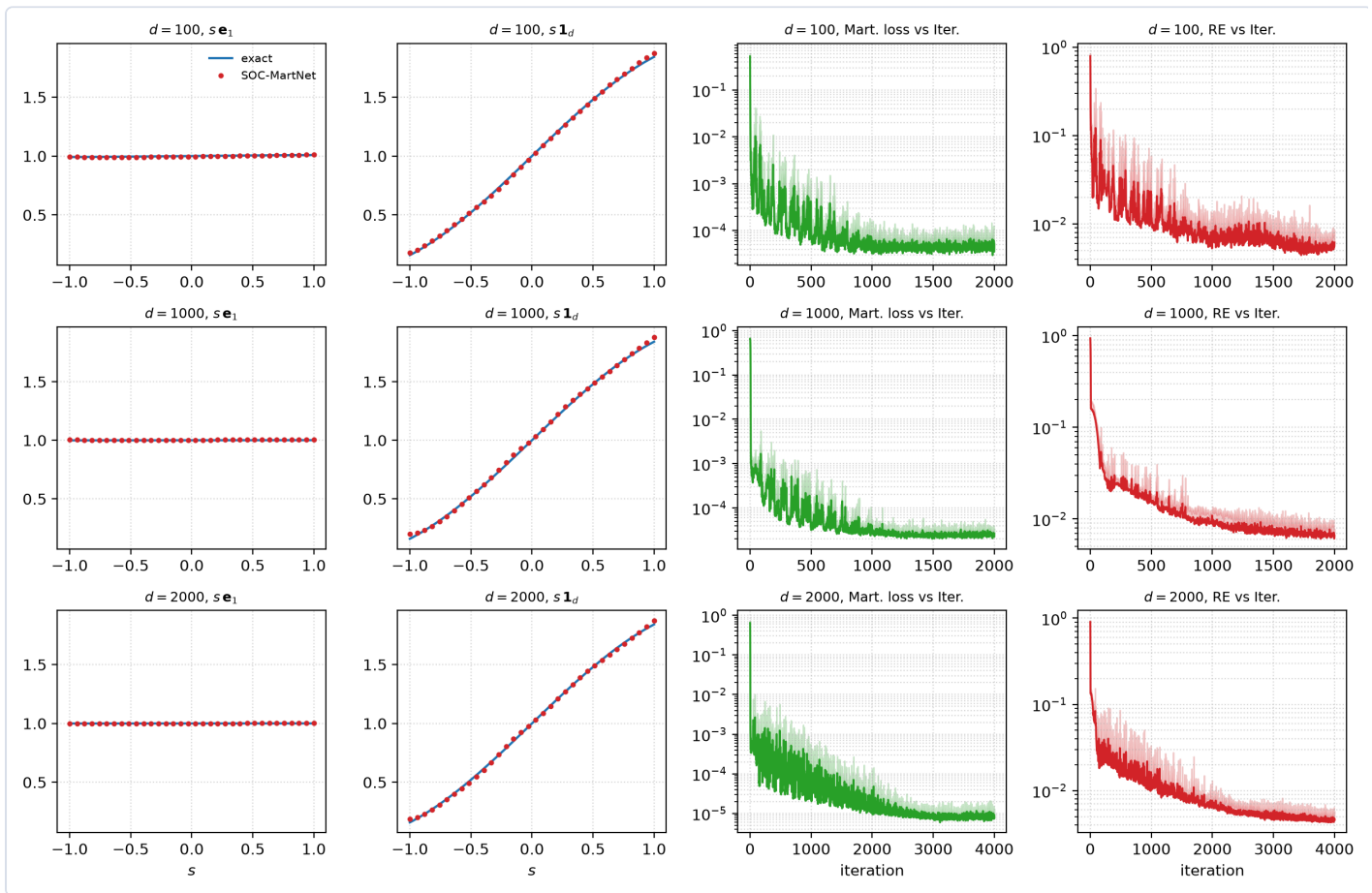


Fig. 1 · §4.1 reproduction (summary data [figures/fig1_linear.csv](#))

Parameter settings: linear equation, $d \in \{100, 1000, 2000\}$, $N=100$ time steps, $I=2000$ (4000 for $d=2000$), $M=10^4$ (6000 for $d=2000$), path renewal 0.2, fp32, seeds 0–4; full command lines in [experiments/sec41_linear_parabolic.sh](#) .

3. §4.2 Semilinear equation (paper Fig. 2)

T	1	0.1	0.05	0.01
RE (5-seed)	1.46e-2 ± 0.27e-2	3.24e-3 ± 0.59e-3	5.64e-3 ± 0.69e-3	2.03e-2 ± 0.51e-3

Hardware: single RTX 4090 (d=100 numerical row); the d=10 reference curves (top row) were produced on an A100.

Paper Fig. 2 (2×4 panels): top row, d=10 exact-solution reference; bottom row, d=100 numerical solution (mean ± 2SD).

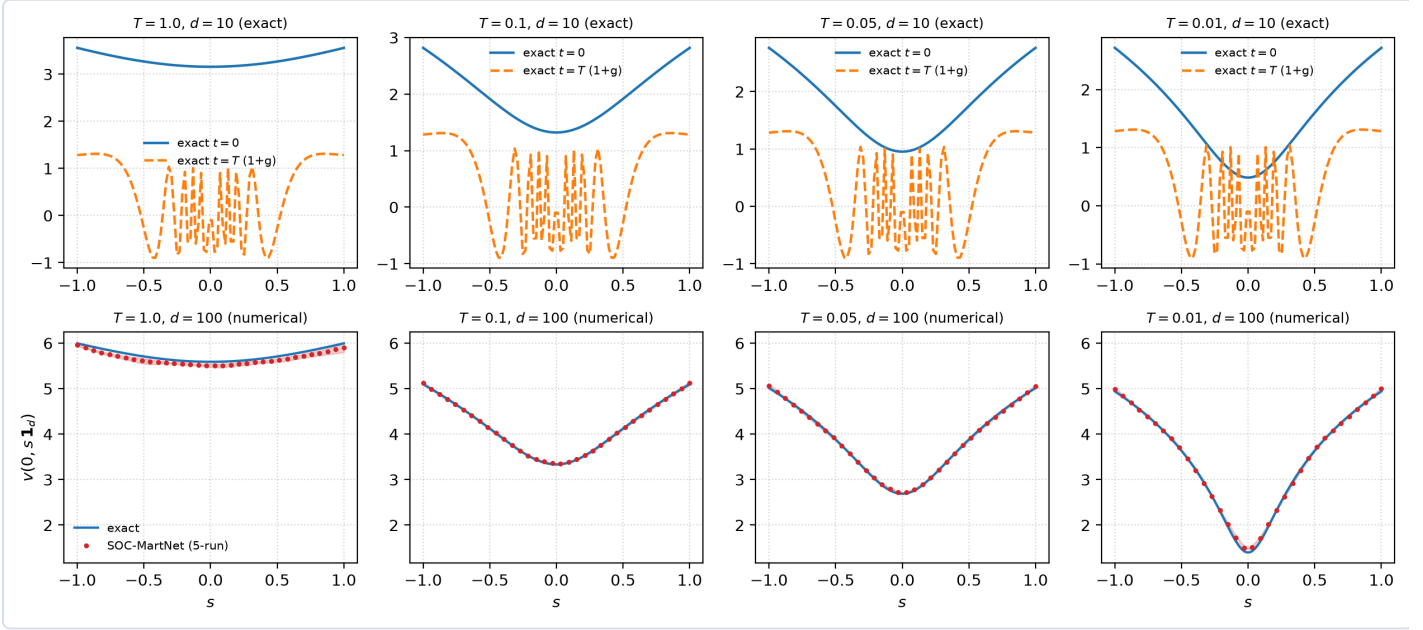


Fig. 2 · §4.2 reproduction (figures/fig2_semilinear.csv)

Parameter settings: semilinear equation, d=100, $T \in \{1, 0.1, 0.05, 0.01\}$, N=100, I=2000, M=10⁴, path renewal 0.2, fp32, seeds 0–4 (top-row reference uses the d=10 analytic convention); full command lines in `experiments/sec42_semilinear.sh`.

4. §4.3 Time-discretization convergence rate (paper Fig. 3)

N	3	4	5	6	8	12	18
Paper anchor	6.44e-2	4.48e-2	3.33e-2	2.66e-2	2.00e-2	1.46e-2	1.03e-2
This reproduction (5-seed)	7.29e-2	5.15e-2	4.51e-2	3.12e-2	2.26e-2	1.65e-2	1.04e-2
Ratio	1.13	1.15	1.36	1.17	1.13	1.13	1.005

Hardware: single RTX 4090 GPU.

The error metric is the paper's notation RE_1 (relative L^1 error). The log–log slope of this reproduction is **−1.087**; the paper's reference slope is $O(N^{-1.01})$. Paper anchor values in the table are displayed to two significant digits; the ratio column is computed from the unrounded anchors.

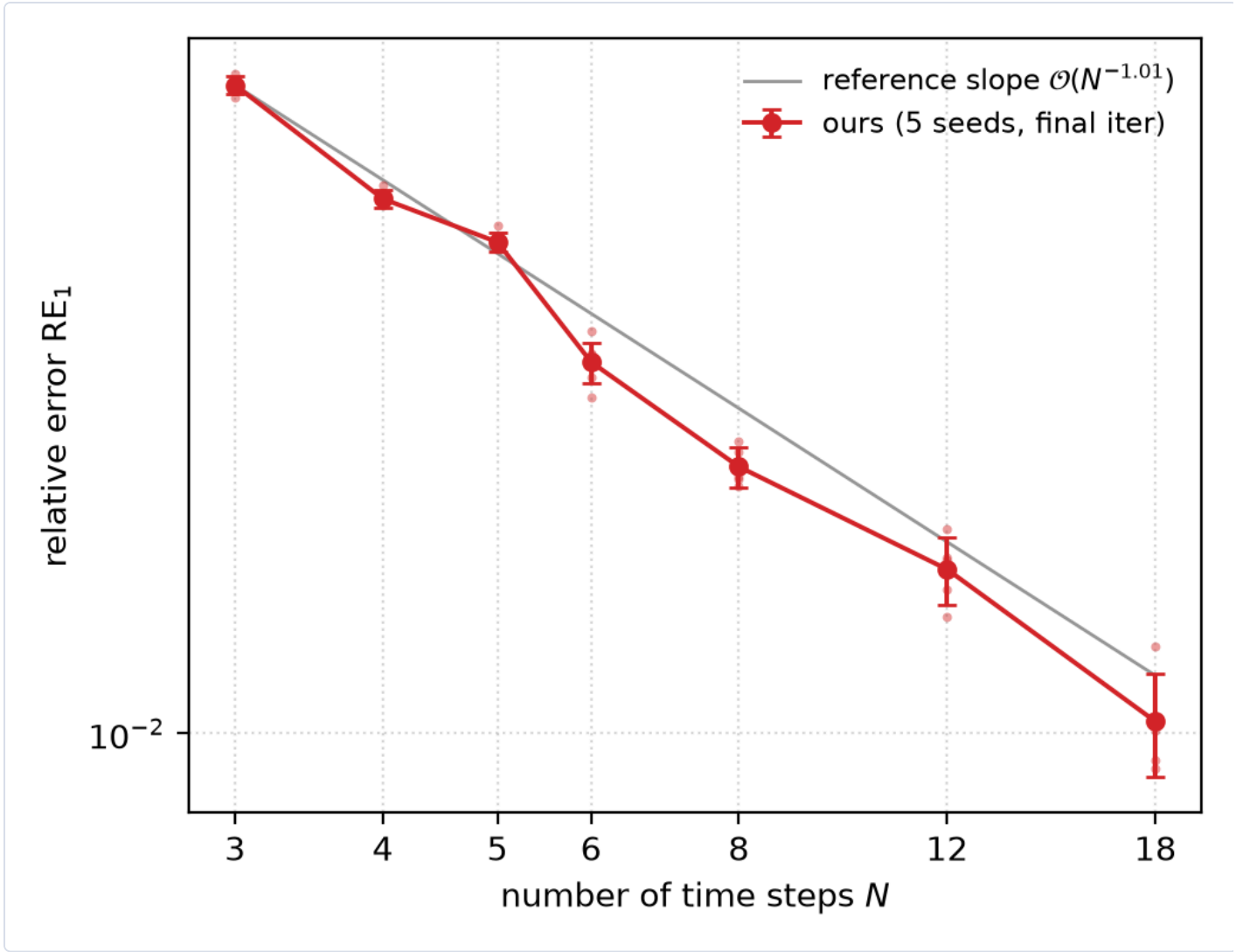


Fig. 3 · §4.3 reproduction (`figures/fig3_time_convergence.csv` ; paper anchor values in the table above)

Parameter settings: LinearSinAC equation, $d=100$, $N \in \{3,4,5,6,8,12,18\}$, $I=3000$ ($N \leq 5$) / 6000 ($N \geq 6$, the paper's two-tier convention), $M=10^4$, path renewal 0.2, fp32, seeds 0–4; full command lines in `experiments/sec43_time_convergence.sh`.

5. §4.4 Non-degenerate HJB (paper Figs. 4 and 5)

5.1 Paper Fig. 4 (HJB-1) 12/12

$d \setminus T$	1	0.5	0.1	0.01
100	6.15e-3	6.44e-3	1.18e-2	3.16e-2
1000	5.48e-3	6.98e-3	1.94e-2	1.14e-1
2000	2.90e-3	2.61e-3	1.96e-2	9.24e-2

Hardware: single A100 GPU.

Parameter settings: HJB-1 with oscillatory terminal ($\varepsilon_0 = 0.1\pi$), $d \in \{100,1000,2000\} \times T \in \{1,0.5,0.1,0.01\}$, $I=1000$, $M=10^4$, path renewal 0.2, fp32, single run (as in the paper's repeat convention); $\bar{\lambda}=1000$ (100 for $d \geq 1000$ with $T=0.01$, per the paper text); full command lines in `experiments/sec44_fig4_hjb1.sh`.

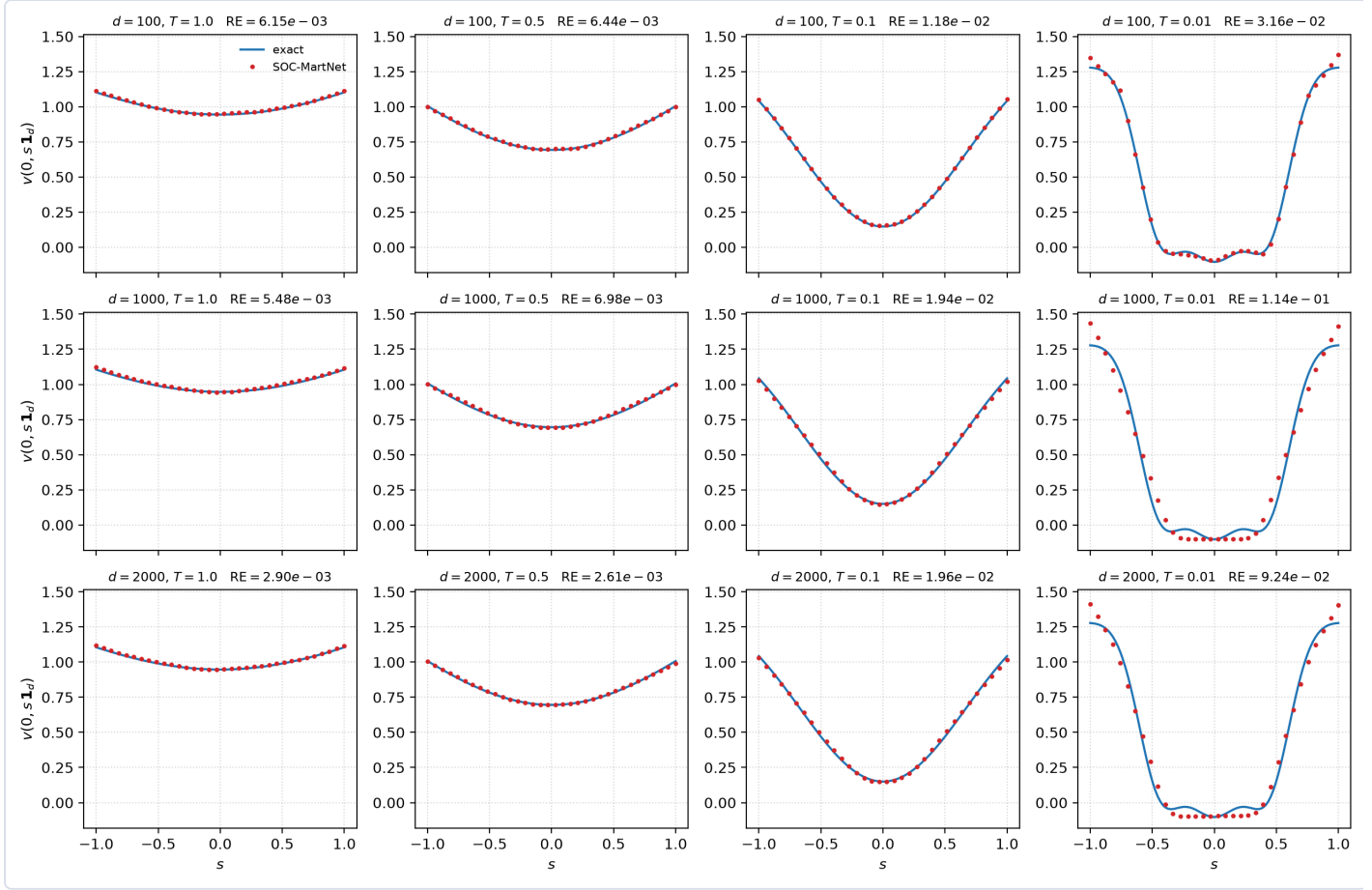


Fig. 4 · reproduction of paper Fig. 4 (`figures/fig4_hjb1_tscan.csv`)

5.2 Paper Fig. 5 (HJB-2/-3) 5-seed ($d=2000$ and $d=10^4$)

Cell	RE (this reproduction)
HJB-2 $d=2000$	0.01757 ± 0.00150
HJB-3 $d=2000$	0.02227 ± 0.00132
HJB-2 $d=10^4$	0.01387 ± 0.00086
HJB-3 $d=10^4$	0.02838 ± 0.00359

Hardware: A100, 8-GPU DDP (all rows).

This subsection of the paper reports illustrative results (no printed numeric anchor); the table above lists the final values of this reproduction. **Protocol notes:** as in paper Fig. 5, every row uses 5 independent runs; the shaded bands are mean $\pm$ 2SD (one-sided on the log axes, per the paper caption). The learning rates, iteration count, and batch size are not printed in the paper; the values used here are given in the parameter settings below.

Parameter settings: $d=2000$ — $I=8000$, $\bar{\lambda}=1000$, $\text{lr}_u=2.744e-5$ and $\text{lr}_\rho=2.744e-4$ ($4\times$ the base), $M=10^4$, seeds 0–4; $d=10^4$ — $I=6000$, $W_h=10010$, batch 64, $\text{lr}_u=7.571e-6$ and $\text{lr}_\rho=7.571e-5$ ($4\times$ the base), $M=10^4$, seeds 0–4. Full command lines in `experiments/sec44_fig5_hjb23.sh`.

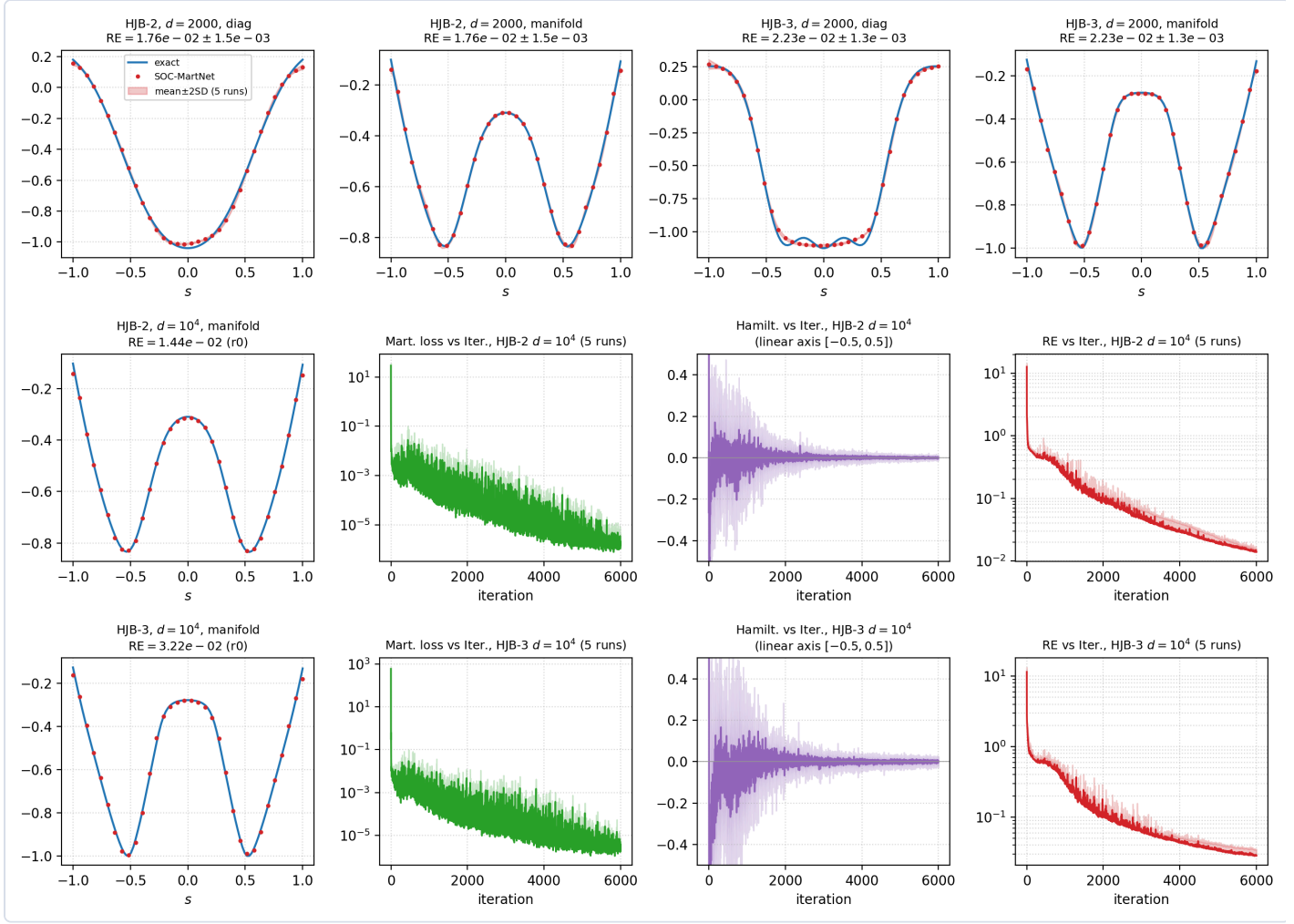


Fig. 5 · reproduction of paper Fig. 5 (3×4 twelve-panel layout matching the paper; $d=2000$ curves show the 5-run mean with mean±2SD bands; Mart/RE on log axes use one-sided mean+2SD bands, per the paper caption;
[figures/fig5_hjb_highdim.csv](#))

6. §4.5 Space-time domain validity (paper Fig. 6)

Metric	This reproduction (5-seed)	Paper reference
RE at end of training	0.02062 ± 0.00172	illustrative result, no printed numeric anchor
Shift-line $ \Delta v $ ($r = 0.125 / 0.25$)	$0.0172 / 0.0308$	—
Path RE, final value	0.0292	—

Hardware: A100 (seeds 1–4) and RTX 4090 (seed 0).

The shift-line $|\Delta v|$ is the mean of $|\Delta v|$ over a 100-point grid on $s \in [0, 1]$ (not the maximum); all values in the table are 5-seed means.

Parameter settings: HJB-2 ($\varepsilon_1 = 0.2$), $d=1000$, $I=8000$, $M=10^4$, path renewal 0.2, $\bar{\lambda}=10$, $lr_u=4.777e-5$ and $lr_\rho=4.777e-4$, fp32, seeds 0–4; the evaluation covers both the shift-line pointwise and the path metrics (`--eval-ptx` / `--eval-path-re`). Full command lines in [experiments/sec45_domain_validity.sh](#) .

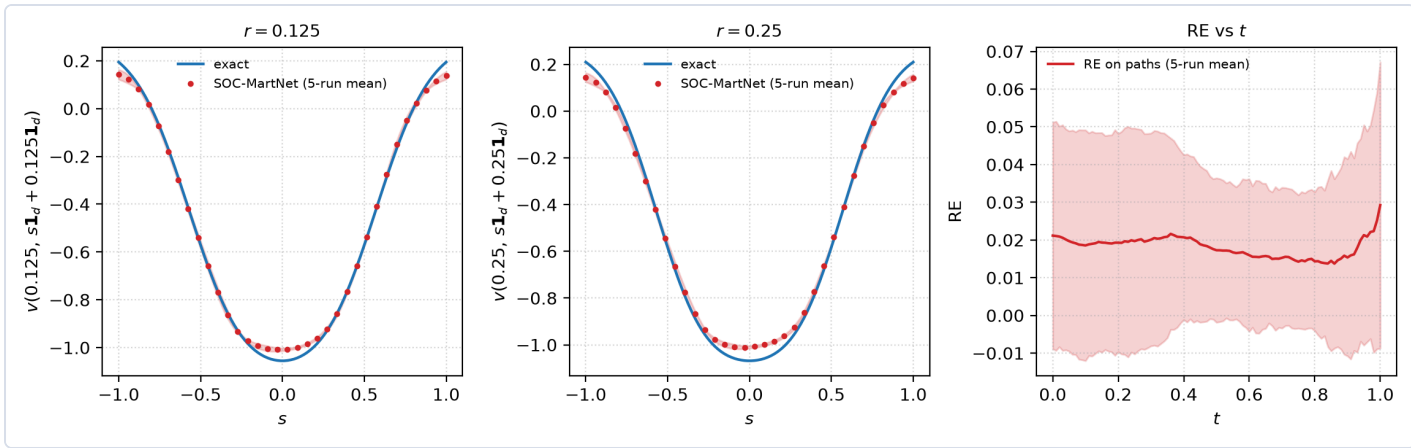


Fig. 6 · §4.5 reproduction (3 panels: shift lines $\times 2$ + RE~t; `figures/fig6_domain_validity.csv`)

7. §4.6 Shifted-terminal SOCP (paper Fig. 7)

Parameter settings: ShiftTarget equation (log-form terminal). Both rows use $I=3000$, $M=10^4$, $\bar{\lambda}=1000$, fp32, seeds 0–4; SD is the sample standard deviation (ddof=1; the summary CSVs under figures/ store the population standard deviation — the two differ by a factor $\sqrt{5/4}$). Full command lines in the §4.6 rows of `experiments.csv` and in `experiments/sec46_shifted_target.sh` .

d	RE (5-seed)	Gap of $J(u)$ to the theoretical lower bound $J(u^*)$
100	$1.45\text{e-}3 \pm 0.44\text{e-}3$	0.11%
500	$3.23\text{e-}3 \pm 0.51\text{e-}3$	0.29%

Hardware: single A100 GPU.

Protocol notes: the hyperparameters used for the two rows differ from the paper text — the paper text prescribes $I=1000$ (a control batch under that setting is retained in `experiments/sec46_shifted_target.sh`), whereas both rows here use $I=3000$; the d=100 row additionally uses strengthened adversarial learning rates ($\text{lr}_u \times 4 = 1.2\text{e-}3$ and $\text{lr}_\rho = 40 \times \text{lr}_u = 4.8\text{e-}2$, path renewal fraction 0.1), while the d=500 row uses the standard convention ($\text{lr}_\rho = 10 \times \text{lr}_u$, renewal fraction 0.2).

Paper Fig. 7 consists of: diag curves + RE history (log axis, one-sided mean+2SD band) + $J(u)$ history (linear axis, mean $\pm$ 2SD); the reproduction is shown in Fig. 7.

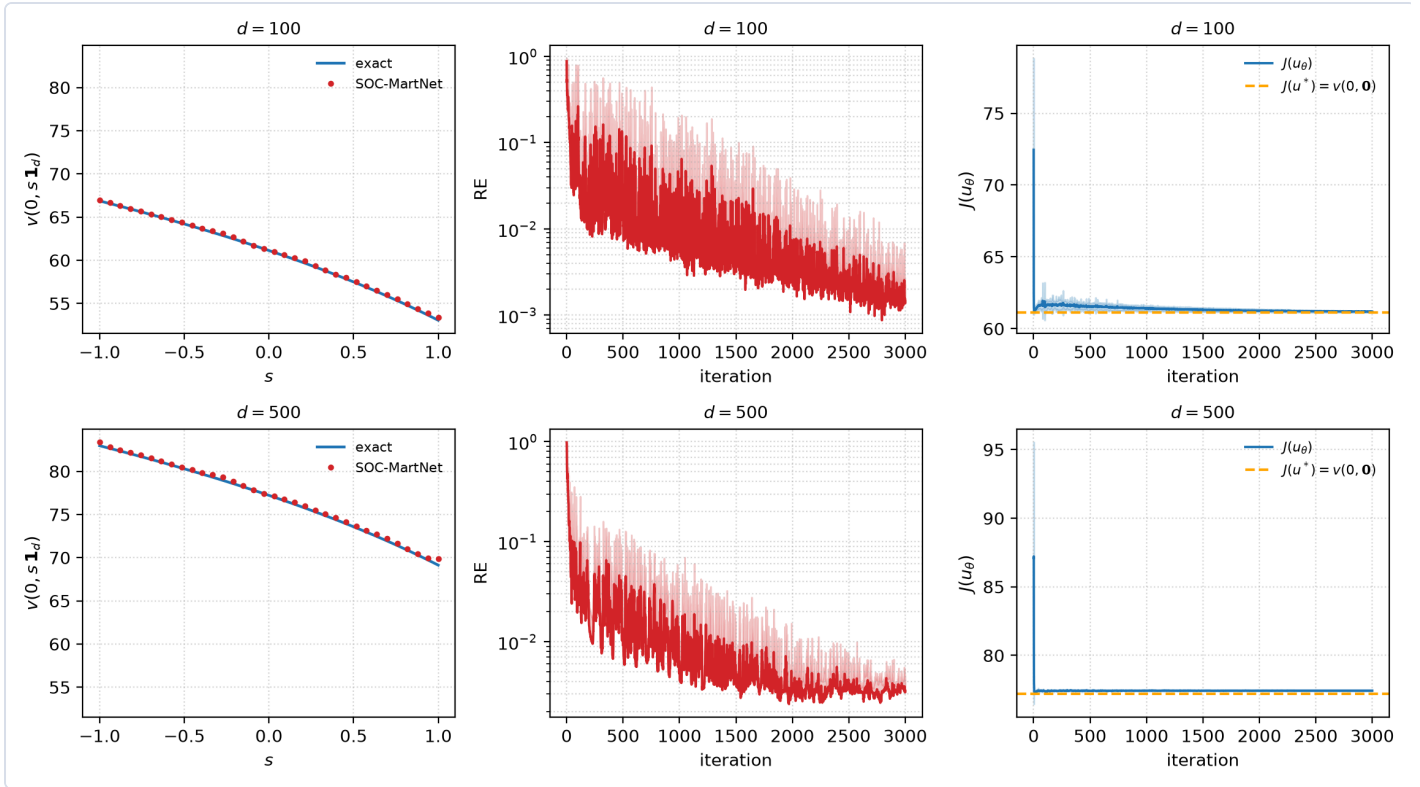


Fig. 7 · §4.6 reproduction (`figures/fig7_shifted_target_summary.csv`)

8. §4.7 ε -perturbation suite (paper Fig. 8)

ε	1	1/2	1/4	1/8
RE vs $\varepsilon=0$ reference (5-seed)	1.743 ± 0.006	0.823 ± 0.005	0.370 ± 0.005	0.203 ± 0.010

Hardware: single A100 GPU.

Paper Fig. 8 shows solution-curve families for decreasing ε (with HJB-2 = $\varepsilon=0$ as reference); in this reproduction the $\varepsilon = \frac{1}{8}$ family has mean -0.569 (reference -0.5).

Parameter settings: HJB-2 ε -perturbation family, $d=1000$, $\varepsilon \in \{1, 1/2, 1/4, 1/8\}$, $I=4000$, $M=10^4$, path renewal 0.2, $\bar{\lambda}=1000$, fp32, seeds 0–4. Full command lines in `experiments/sec47_epsilon.sh`.

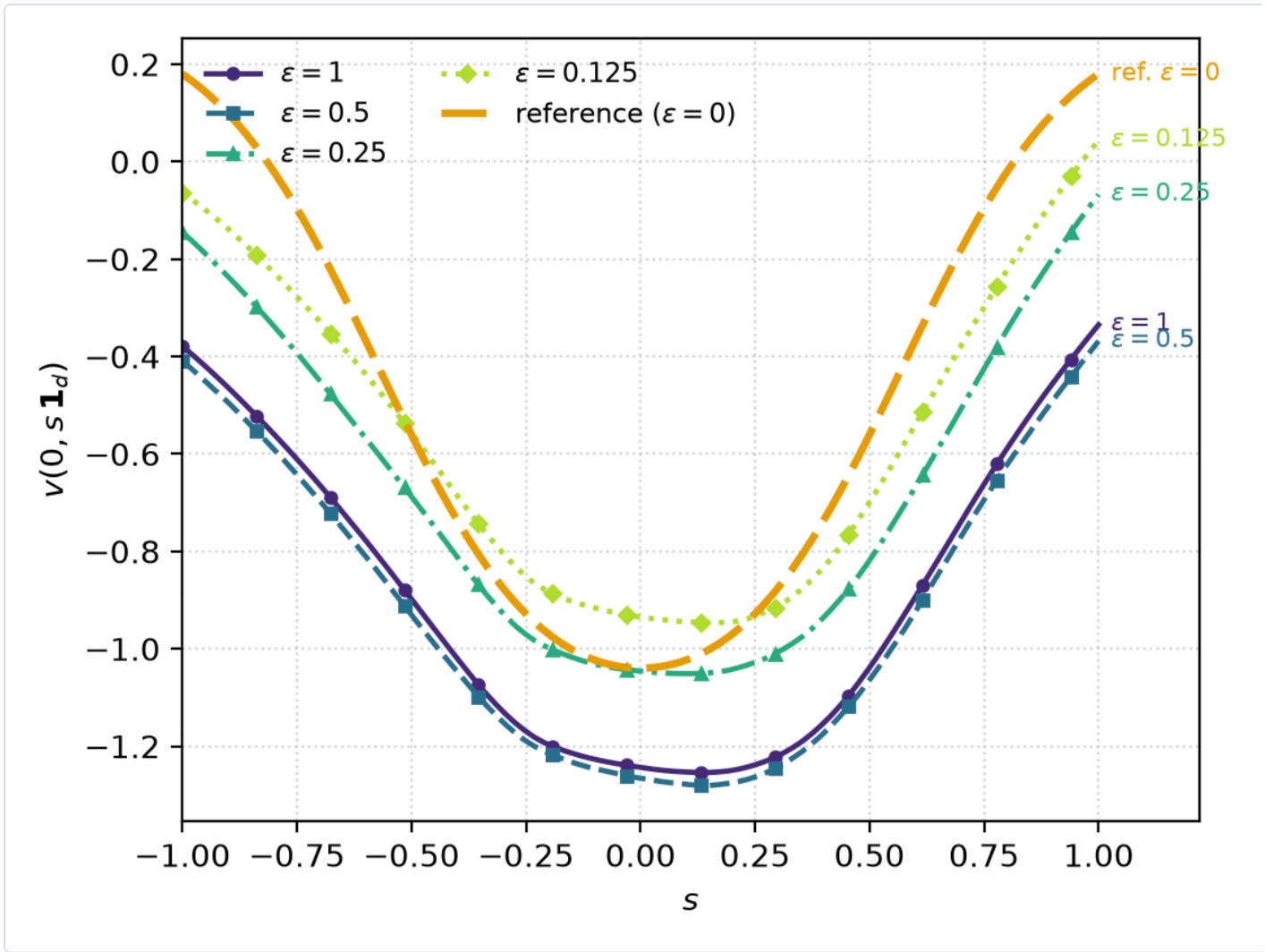


Fig. 8 · §4.7 reproduction (`figures/fig8_epsilon.csv`)

9. Paper Table 1 (§4.8): equal-time reproduction

Final reproduction protocol = equal-time comparison: MartNet is run within the runtime budget of the DeepBSDE entries reported in paper Table 1 (RT column), after which the two errors are compared. Network and hyperparameters follow the paper text; the iteration count I of each cell is the time budget divided by the per-iteration cost measured with this package.

Cell	I	RE, this reproduction (5-seed)	DeepBSDE printed	Ratio	RT (s), this reproduction	DeepBSDE RT (s), printed
d100 w256	3200	$1.67\text{e-}3 \pm 4.71\text{e-}4$	$3.23\text{e-}3$	0.52×	72.3	73
d300 w256	3300	$6.50\text{e-}4 \pm 7.80\text{e-}4$	$1.18\text{e-}3$	0.55×	89.2	90
d500 w256	4600	$1.07\text{e-}3 \pm 6.24\text{e-}4$	$1.05\text{e-}3$	1.02×	129.3	116

Cell	I	RE, this reproduction (5-seed)	DeepBSDE printed	Ratio	RT (s), this reproduction	DeepBSDE RT (s), printed
d800 w256	4800	2.36e-3 ± 3.65e-3	2.54e-3	0.93×	155.4	145
d1000 w256	4900	9.23e-4 ± 5.62e-4	1.86e-3	0.50×	169.9	170
d100 wd10	3500	1.79e-3 ± 1.59e-3	2.86e-3	0.63×	52.1	53
d300 wd10	3500	5.12e-4 ± 4.02e-4	3.27e-4	1.57×	87.2	103
d500 wd10	3500	1.98e-3 ± 1.36e-3	6.41e-4	3.09×	167.6	184
d800 wd10	3000	2.43e-3 ± 1.61e-3	1.28e-3	1.90×	328.9	386
d1000 wd10	4100	1.47e-3 ± 1.07e-3	3.77e-3	0.39×	603.6	615

Hardware: single A100 GPU (both the reproduction and the paper's reported runs).

Ratio = 5-seed mean of this reproduction / DeepBSDE printed value. RT = cumulative training wall-clock time (rt column of the training CSV), 5-seed mean; the printed DeepBSDE RT is the time budget of this protocol (the paper's reported single-A100 runtime). Realized runtimes are 85%–111% of the corresponding budgets. Per-seed values are in `results/t1e_final_table.csv`. The largest error ratio in the table is 3.09×, the geometric mean is 0.89×, and 6/10 cells are below 1×.

Protocol notes: deviations from the printed parameters — (i) iteration count I: the paper Table 1 caption states I=1000, whereas this reproduction uses 3200–4900 per the equal-time budget (derivation in the intro of this section); (ii) lr_ρ : the paper text prescribes a flat 1e-2; the five low-dimensional cells here keep 1e-2, and the five high-dimensional cells use the relative convention $10 \times \text{lr}_u$ (lr_u for the w256 d1000 cell); (iii) batch size: the paper §4 opening tiers it by dimension as 256/128, whereas this reproduction uses 128 uniformly. Per-cell values in `results/t1e_final_table.csv`.

Parameter settings: HJB-2 ($\epsilon_1 = 0.2$), point evaluation $x_0=0$, networks with $h=4$ (paper §4.8 text: “four hidden layers, W_h units”), $W_h \in \{256, d + 10\}$, rho network $r=600$ (paper default), batch=128 (uniform across the table), $\text{lr}_u = 3 \times 10^{-3} d^{-0.5}$ (paper §4 opening learning rate), $\bar{\lambda}=1000$, $M=10^4$, path renewal 0.2, fp32, seeds 0–4, single GPU; per-cell lr_ρ and I in `results/t1e_final_table.csv`. Full command lines in `experiments/table1_equaltime.sh` (50 jobs).

10. §4.8 Paper Table 2 (DDP runtimes) 20/20

The reproduction protocol uses a **uniform batch size of 128** (identical across all 20 cells, making the table comparable across dimensions and GPU counts).

Parameter settings: HJB-3 ($\epsilon_1 = 0.1$), $W_h = d + 10$, I=6000, $M=10^4$, path renewal 0.2, $\bar{\lambda}=1000$, fp32, single run (as in the paper's repeat convention), GPU counts {1,2,4,8} (DDP), batch = 128. Full command lines in the Table-2 rows (uniform 128) of `experiments/experiments.csv` and in `experiments/table2_runtime.sh`.

GPUs \ d	100	500	800	1,000	2,000
1 × A100	111.4 (1.0×)	409.7 (1.0×)	961.0 (1.0×)	1299.5 (1.0×)	4811.7 (1.0×)
2 × A100	136.1 (0.8×)	230.4 (1.8×)	517.5 (1.9×)	701.3 (1.9×)	2500.8 (1.9×)
4 × A100	150.4 (0.7×)	151.2 (2.7×)	295.5 (3.3×)	387.6 (3.4×)	1354.1 (3.6×)
8 × A100	152.1 (0.7×)	155.6 (2.6×)	194.1 (5.0×)	235.3 (5.5×)	773.8 (6.2×)

Hardware: A100 (DDP with the GPU count listed per row; the printed values are the paper's reported A100 runs).

The table layout matches paper Table 2 (rows = GPU count, columns = dimension). Format: RT in seconds (parenthesized value = DDP speedup = $rt(g_1)/rt(g_k)$); RT = cumulative training wall-clock time (rt column of the training CSV); the printed values and this reproduction both ran on A100 GPUs. Full data including the RE column in `results/t2b_fix_agg.csv` (per-run final-row extract: `results/t2b_raw_final.txt`). Cell-by-cell ratios against the paper's printed values: **0.53–1.06** (0.96–1.00 for the d2000 column). DDP speedup pattern: 2.6×–6.2× on 8 GPUs for d500–d2000, flat for d100.

11. Reproduction guide (using this package)

This directory is **self-contained**: source code + command lines of all experiments + reference values + figures and plotting scripts. Directory layout:

```

repository root (= the workdir)/
├─ REPORT.html           # this report
├─ REPORT.pdf            # print version of this report (regenerate:
plots/build_report_pdf.py)
├─ code/SOCMartNet-v3-refactored/
|                           # reproduction code: entry run.py + the socmartnet package +
tests/validation_anchors/
├─ slurm/                 # frozen-copy submission channel (submit_job.sh + run_one.slurm)
├─ experiments/
|   └─ experiments.csv    # per experiment family: section / hardware / seeds /
representative jobid / command-line template
|   └─ table1_equaltime.sh ... table2_runtime.sh
|   |                           # 10 single-case reproduction scripts (run on demand; each header
states the paper content covered, the reference values, and the expected CSV line count)
|   └─ smoke_test.sh      # quick end-to-end check (prints SMOKE OK)
|   └─ common.sh          # shared harness (WORKDIR / submit()); each single-case script
sources it)
|   └─ reproduce_all.sh   # full driver (invokes the 10 single-case scripts in order;
parameters taken verbatim from the job-log headers that produced the results)
├─ results/               # reference values: Table1/Table2/$4.1-4.3/$4.4 aggregate CSVs +
final-row extracts of this reproduction
|   └─ final_rows.csv     # final-row extract of the runs behind the shipped results (jobid /
stem / line count / final values)
├─ katex/                 # local KaTeX rendering library (formula rendering, works offline)
└─ figures/               # figure PNGs + per-figure summary-data CSVs (pointwise curve data

```

```
live in the archived job CSVs)
└─ plots/                               # plotting and aggregation scripts (plot_*.py / build_*.py)
```

Reproduction steps (any A100-class SLURM cluster; the numbers in this package were produced on A100 and RTX 4090 GPUs):

```
# 1) This repository doubles as the workdir: clone it on your cluster and work at its
#    root -- submit_job.sh expects exactly this layout (code/SOCMartNet-v3-refactored/
#    + slurm/ + experiments/). Adapt the site template slurm/run_one.slurm:
#    #SBATCH account/partition + PY variable
# 2) Submit: run the single-case scripts on demand (experiments/table1_equalitytime.sh ...
#    table2_runtime.sh),
#    or bash experiments/reproduce_all.sh for the full suite; below, the §4.4 paper-Fig. 5 d1e4
#    formal batch as an example
export SBATCH_PARTITION=batch SBATCH_GPUS=8 SBATCH_TIMELIMIT=1-00:00:00
bash slurm/submit_job.sh fig5_hjb2_d1e4_s1 \
  --example hjblq --dim 10000 --delta0 0.2 --preset sisc --dtype float32 \
  --max-iter 6000 --num-paths 10000 --renew-frac 0.2 --fd-residual \
  --lam-bar 1000 --controlled-pool --width 10010 \
  --batch-size 64 --lr-uv 7.5714866e-06 --lr-rho 7.5714866e-05 \
  --seed 1 --tag fig5_hjb2_d1e4_s1
# 3) Success check: sacct COMPLETED alone is not sufficient; the CSV line count is authoritative
#    (6000 iterations → 6002 lines)
# 4) Compare against reference values: aggregate CSVs and final_rows.csv under results/
```

Environment: Python 3.12 everywhere; PyTorch 2.11.0+cu128 (A100 cluster) / 2.6.0+cu124 (RTX 4090 cluster); RMSprop; fp32; multi-GPU DDP. Accuracy sections use ≥ 5 seeds (0–4) per cell; the single-run exceptions are paper Table 2 and all 12 cells of paper Fig. 4.